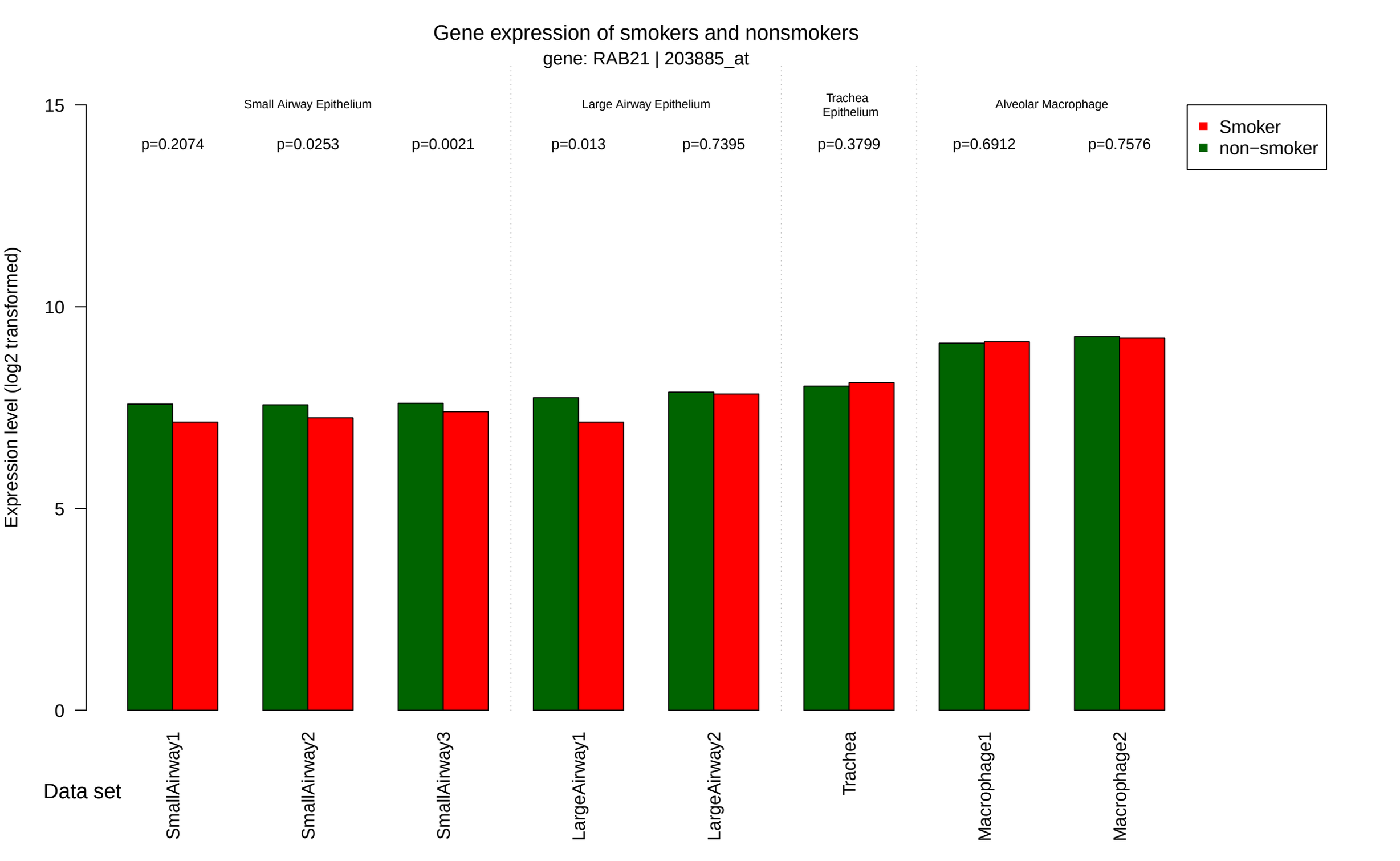
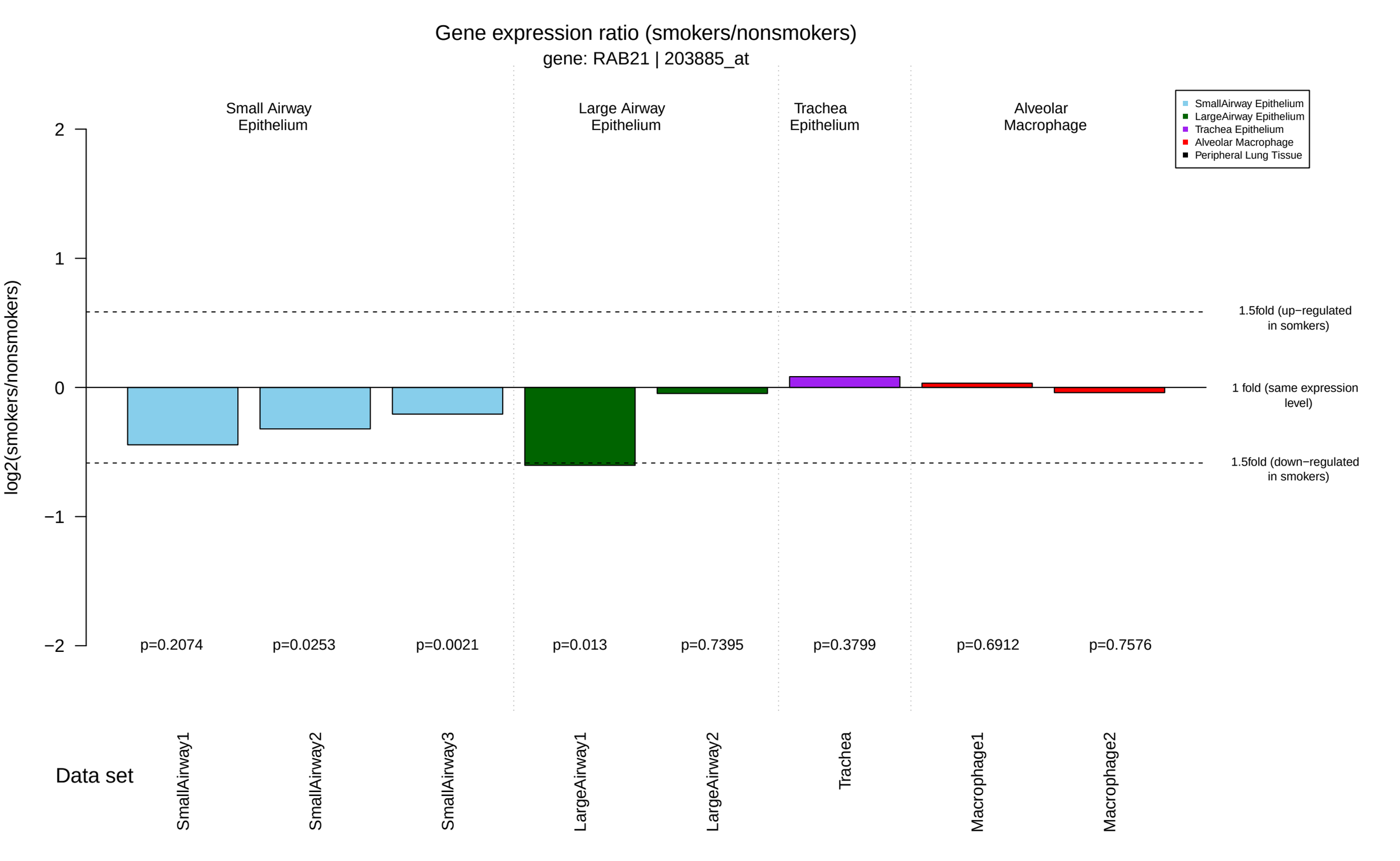
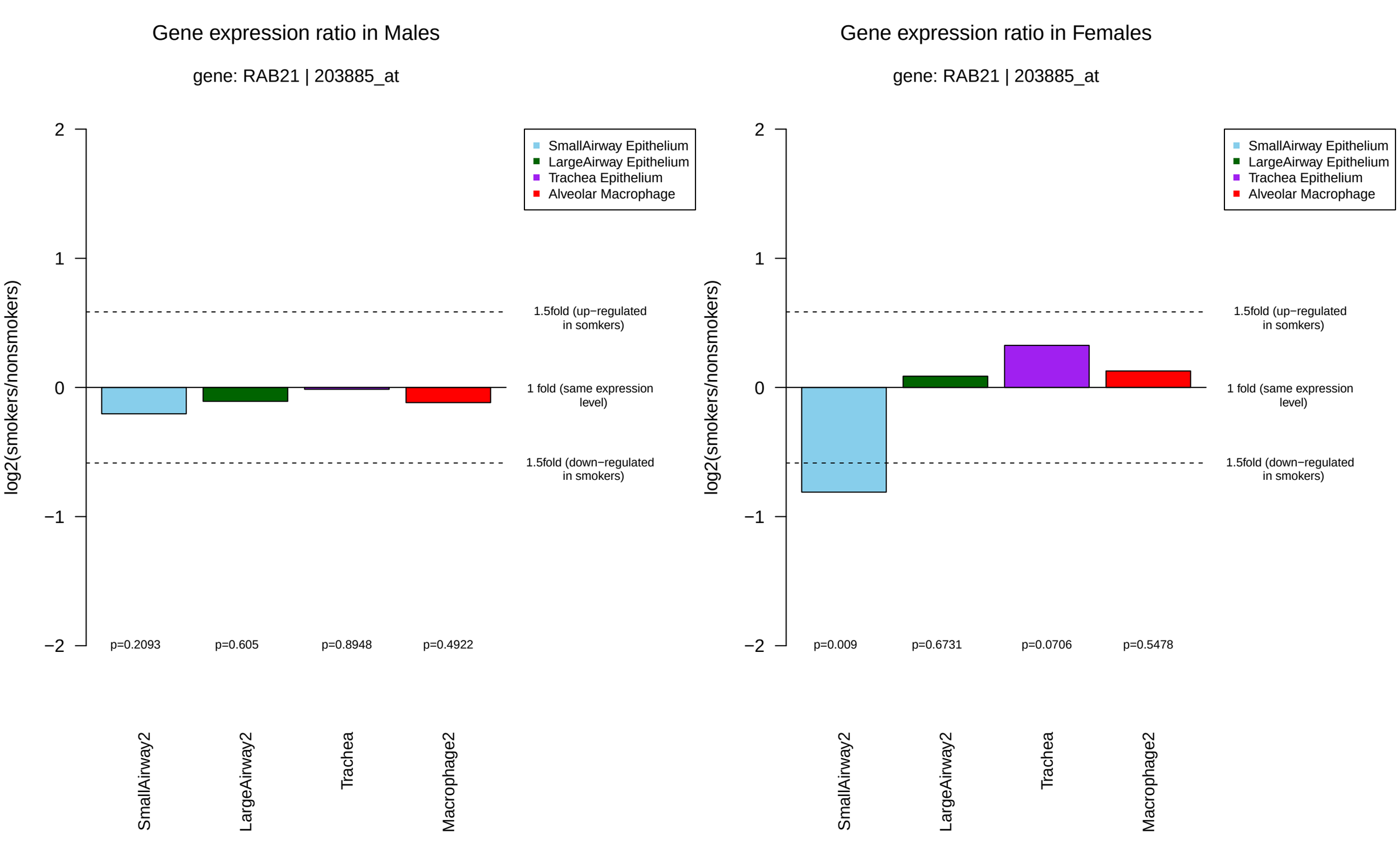
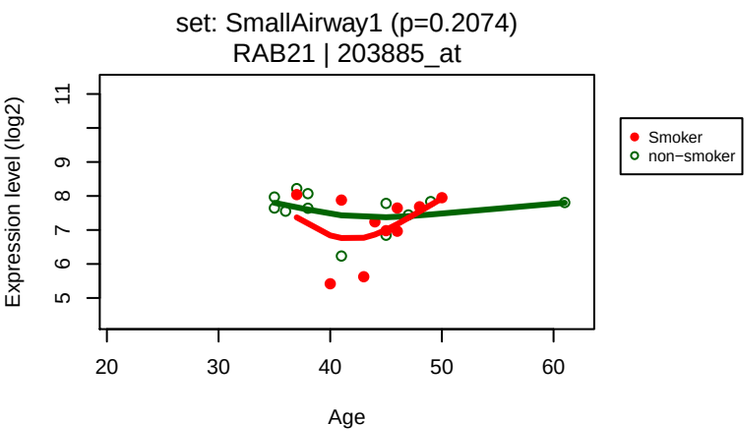
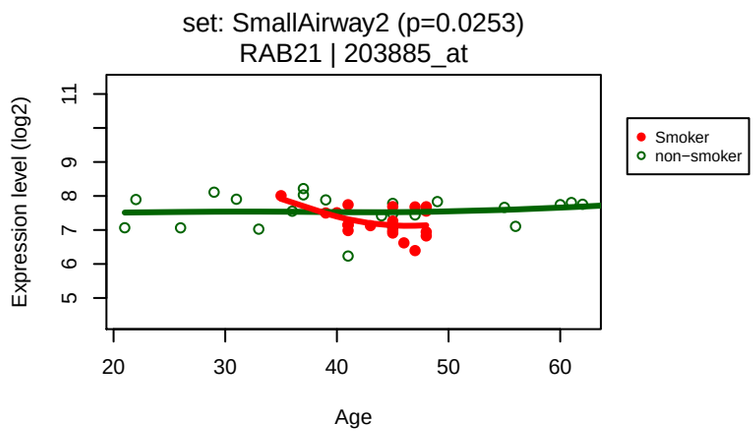
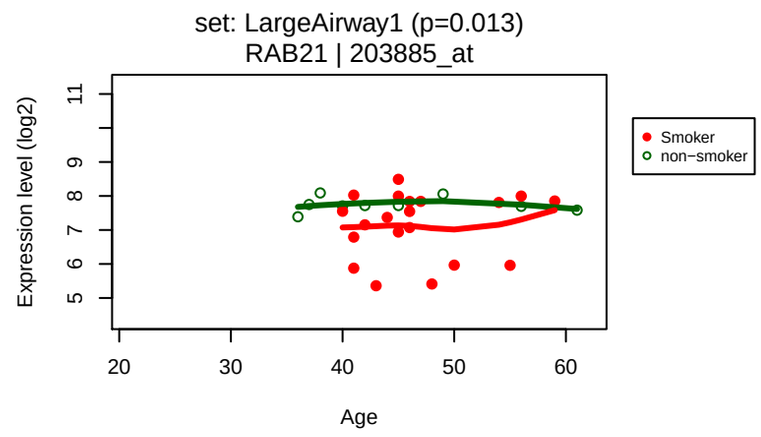

Gene expression of smokers and nonsmokers
gene: RAB21 | 203885_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2074
p=0.0253
p=0.0021
p=0.013
p=0.7395
p=0.3799
p=0.6912
p=0.7576
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

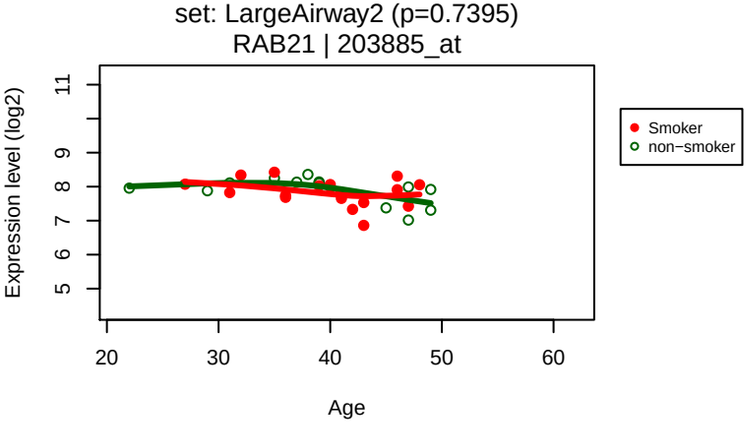

Gene expression ratio (smokers/nonsmokers)
gene: RAB21 | 203885_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2074
p=0.0253
p=0.0021
p=0.013
p=0.7395
p=0.3799
p=0.6912
p=0.7576
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

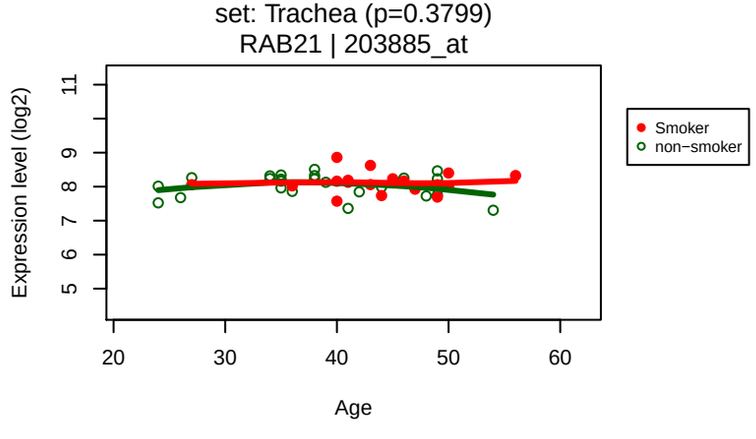

Gene expression ratio in Males
gene: RAB21 | 203885_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2093
p=0.605
p=0.8948
p=0.4922
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB21 | 203885_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.009
p=0.6731
p=0.0706
p=0.5478
SmallAirway2
LargeAirway2
Trachea
Macrophage2

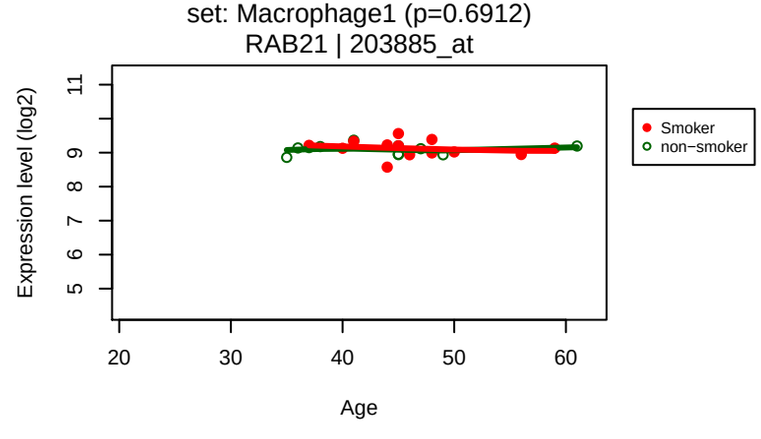

set: SmallAirway1 (p=0.2074)
RAB21 | 203885_at
Expression level (log2)
Age
Smoker
non−smoker

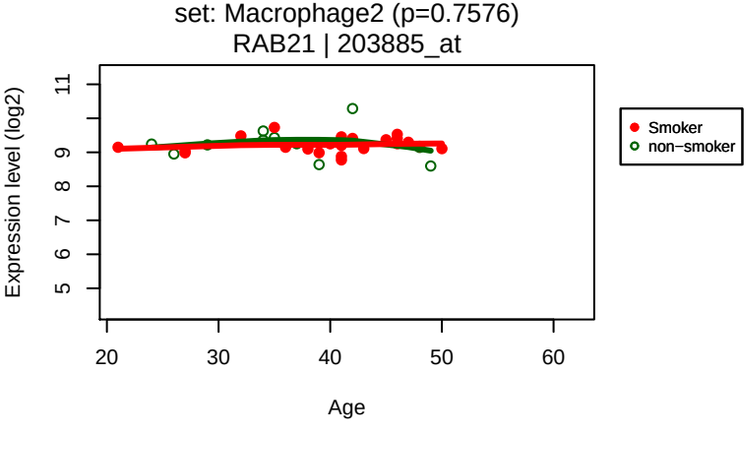

set: SmallAirway2 (p=0.0253)
RAB21 | 203885_at
Expression level (log2)
Age
Smoker
non−smoker

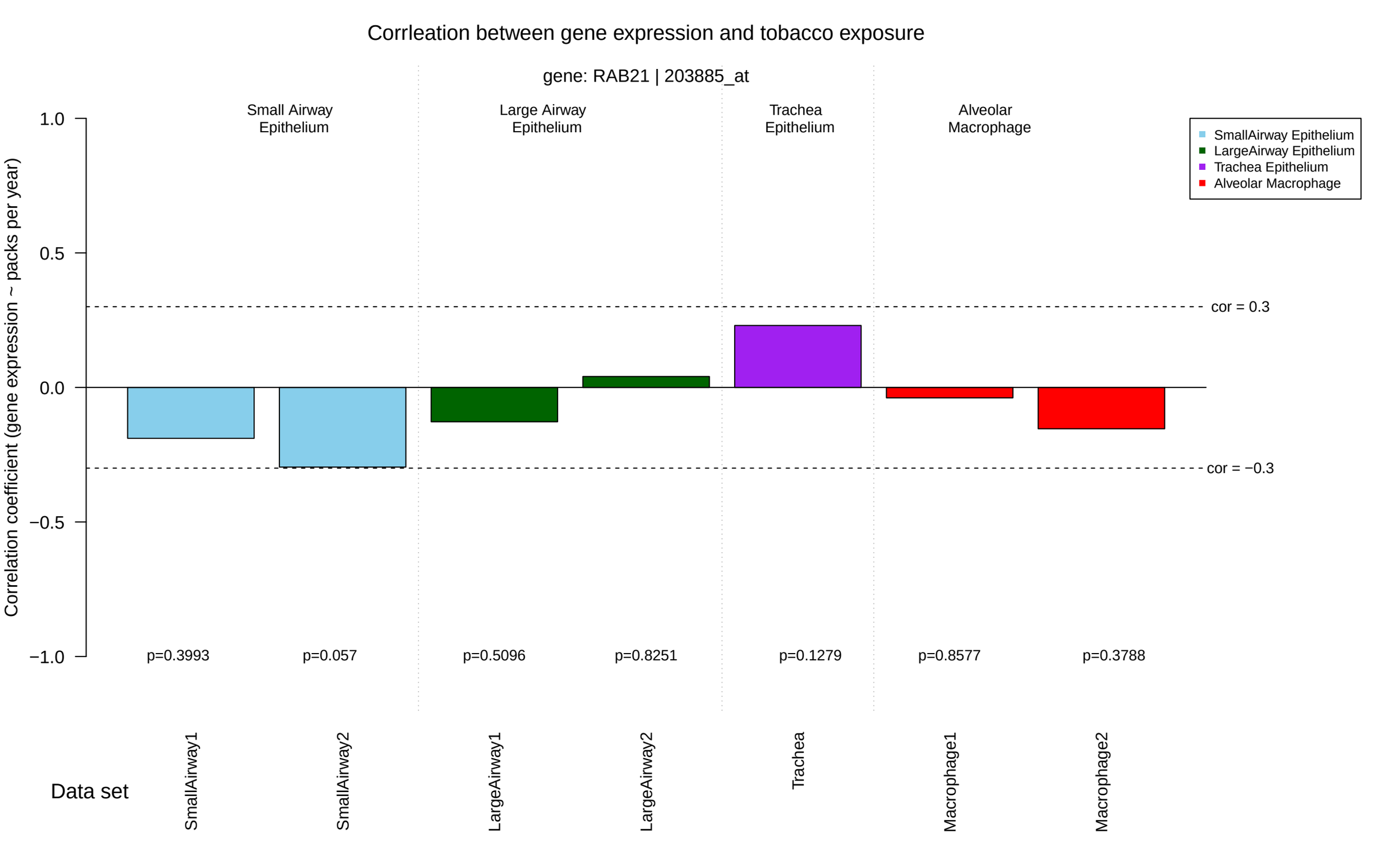

set: LargeAirway1 (p=0.013)
RAB21 | 203885_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7395)
RAB21 | 203885_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3799)
RAB21 | 203885_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6912)
RAB21 | 203885_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7576)
RAB21 | 203885_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB21 | 203885_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3993
p=0.057
p=0.5096
p=0.8251
p=0.1279
p=0.8577
p=0.3788
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2